

AI STUDY ENGINE

Software Design Document (SDD) - Hackathon Edition

1. Executive Summary

AI Study Engine is a unified student operating system that combines personalized AI mentorship, resource management, internship recommendations, career guidance, productivity management, study assistance, analytics, and opportunity discovery into a single platform.

2. Problem Statement

Students currently rely on multiple disconnected platforms for learning, notes management, career guidance, internships, productivity, and opportunities. This creates fragmentation, poor organization, and inefficient learning experiences.

3. Vision

Create a centralized AI-powered ecosystem where students can learn, organize resources, discover opportunities, receive personalized guidance, and track growth.

4. User Personas

• First Year Students • Placement-Oriented Students • Competitive Programming Enthusiasts • AI/ML Learners • Working Professionals Upskilling

5. System Modules

Dashboard AI Mentor Knowledge Vault Roadmaps & Guides Internship Hub Resume Studio Study Assistant Productivity Hub Opportunity Center Analytics Profile & Personalization

6. Dashboard Design

Widgets: • Learning Progress • Recent Resources • Recommended Internships • Upcoming Deadlines • AI Suggestions KPIs: Study Streak, Tasks Completed, Applications Sent, Resources Saved.

7. AI Mentor Architecture

Modes: Career Coach Study Coach Resume Coach Interview Coach Productivity Coach Data Sources: User Profile + Vault + Roadmaps + Activities + Applications Outputs: Personalized recommendations and action plans.

8. Knowledge Vault Architecture

Supported Content: PDFs, Images, Links, Notes, Snippets. AI Features: Auto Summary Keyword Extraction Topic Classification Flashcard Generation MCQ Generation Storage: MongoDB Metadata + Cloud Storage.

9. Roadmaps & Guides Engine

Input: Learning Topic Output: Beginner → Intermediate → Advanced → Projects → Interview Preparation → Internship Readiness Examples: Frontend Development AI/ML Cyber Security Data Science DSA

10. Internship Recommendation Engine

Inputs: Branch Skills Interests Location Preference Career Goal Approach: TF-IDF + Cosine Similarity Outputs: Match Score Skill Gap Analysis Recommended Opportunities

11. Resume Studio

Functions: Resume Analyzer ATS Checker Resume Builder Project Recommendations Outputs: Resume Score ATS Compatibility Improvement Suggestions

12. Study Assistant

Upload Notes/PDFs. Generate: Summary Quiz MCQs Flashcards Revision Notes Important Questions

13. Productivity Hub

Manage: Tasks Goals Habits Deadlines AI Recommendations: Weekly Planning Missed Tasks Time Management Suggestions

14. Opportunity Center

Categories: Internships Hackathons Coding Contests Scholarships Workshops Certifications

15. Analytics Module

Track: Study Hours Skills Acquired Applications Submitted Roadmaps Completed Generate AI Insights.

16. Database Design

Collections: Users Documents Roadmaps Internships Applications Tasks Analytics Chat History Opportunities

17. API Design

/login /register /upload /chat /generate-roadmap /recommend-internships /analyze-resume /generate-summary /get-opportunities

18. Technology Stack

Frontend: React + Tailwind Backend: FastAPI Database: MongoDB AI: Gemini API LangChain Scikit-Learn Hosting: Vercel + Render

19. System Workflow

User Login ↓ Profile Creation ↓ AI Personalization ↓ Knowledge Storage ↓ Roadmap Generation ↓ Internship Recommendations ↓ Continuous Analytics

20. Hackathon MVP

Must Build: Dashboard AI Mentor Knowledge Vault Roadmaps & Guides Internship Hub Resume Studio Optional: Study Assistant Analytics Opportunity Center

21. Future Scope

Voice AI Mentor College ERP Integration AI Mock Interviews Placement Predictor Collaborative
Study Rooms Learning Communities Mobile Application

22. Team Distribution

Frontend Developer: Dashboard + UI Backend Developer: API + Database AI Developer: Mentor +
Recommendation Engine Presentation Lead: Documentation + PPT + Demo

23. Judging Points

Innovation Practical Impact Scalability AI Integration Student Personalization Problem Relevance

24. Final Sidebar Structure

Dashboard AI Mentor Knowledge Vault Roadmaps & Guides Internship Hub Resume Studio Study
Assistant Productivity Hub Opportunity Center Analytics Profile